

OPERATOR GUIDE

PDW v5.5.1

Paging Decoder for Windows

2026 Release - Public Beta 2

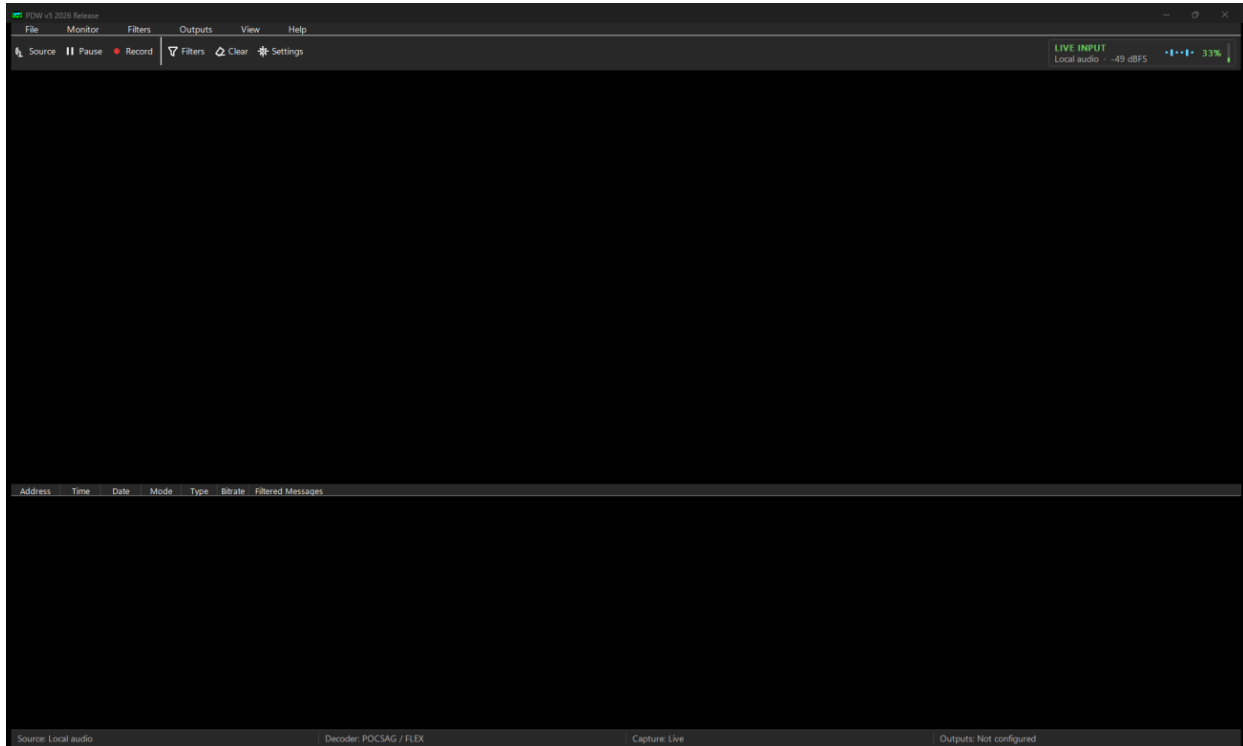


Figure 1. Content-free PDW v5 command bar and live-input layout.

Edition 2.0 | 11 August 2026 | Covers PDW v5.5.1 Public Beta 2

Native x64 and Win32 applications for supported Windows 10/11 and Windows Server systems

All addresses, messages, accounts, endpoints and credentials in this guide are synthetic or masked.

How to use this guide

Training data: Use only synthetic, redacted or explicitly licensed examples for setup, screenshots and acceptance. Never place private pager traffic, recordings, credentials, Windows endpoint IDs or operator settings in a public report.

This manual replaces the 2010 PDW v3.1 guide and the interim v4.1 draft. It keeps the established protocol, decoder, filter and logging concepts, but follows the v5.5.1 installer, dual-architecture packages, 2026 command bar, Settings Center, Capcode Directory, message history, secure outputs and named local-audio profile.

Feature status used in this manual

Status	Meaning
Public Beta 2	Included in the published v5.5.1 prerelease. It remains unsigned and is not a signed stable release.
Preserved legacy	Established decoder, serial, slicer, WinMM, filter or display behaviour intentionally retained for compatibility.
Optional / off	Included but disabled until an operator deliberately enables and configures it.
Hardware-unverified	Automated checks pass, but representative physical-device acceptance remains incomplete.
Development preview	Visible in the active 11 August 2026 source worktree, but not part of the published Public Beta 2 contract.

Ten-minute first run

1. Download the v5.5.1 Public Beta 2 Setup only from the maintained GitHub release and verify its published SHA-256 checksum.
2. Run Setup, read the unknown-publisher warning, and choose x64 for normal current Windows use or Win32 only when a legacy 32-bit receiver or integration requires it.
3. Choose Standard PDW settings. Select the Adelaide FLEX profile only for the documented SDR# and VB-Audio Cable workflow.
4. Start PDW, open Settings, and choose the intended source under Signal & radio. Use only transmissions you are authorised to receive.
5. Select POCSAG/FLEX, ACARS, MOBITEK or ERMES and confirm the live meter responds without persistent clipping.
6. Open Filters > Capcode Directory and create one synthetic training rule. Keep external output routing off for the first test.
7. If local history is required, enable it under View > Message history and decide separately whether message text may be retained.
8. Run configuration-only tests for any approved email, notification, transfer, publishing or data-output destination before enabling live delivery.
9. Create an encrypted Settings > General > Backup / Restore file after configuration and protect its password separately.

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1. Read this first

What PDW does

PDW is a native Windows application that receives, decodes, displays, filters, logs and optionally routes supported digital radio messages. Current development is deliberately additive: modern Windows input, user-interface and output features sit beside established decoder and hardware paths rather than replacing them.

Protocol	Supported role	Compatibility boundary
POCSAG	512, 1200 and 2400 baud	Established parser and serial/audio routes retained.
FLEX	1600, 3200 and 6400 baud	Established parser retained; optional fragment assistance is separately gated.
ERMES	6250 baud	Established decoder route retained.
ACARS	Aircraft data messages	Parity and label/database options remain operator controlled.
MOBITEX	8000 baud in normal use	Established base/mobile and message controls retained.

What Public Beta 2 includes

Area	Included capability	Default / evidence boundary
Windows	Guided Setup, portable packages, x64 and Win32, light/dark UI and modeless Settings Center.	Unsigned beta; full DPI, keyboard, High Contrast and physical-radio matrices remain open.
Signal	WinMM, exact-endpoint WASAPI, serial/slicer, RTL-TCP, RTL-SDR, WAV/SigMF record and replay.	Named Adelaide profile is opt-in and hardware-unverified.
Local tools	Capcode Directory, optional SQLite history, CSV export, local dashboard and isolated extra receiver workers.	History, dashboard and extra channels are off until enabled.
Routing	Email, Apprise, FTPS/SFTP, publishing, MQTT, SQLite, MySQL/ODBC, Telnet and Windows notifications.	Every network/output path is disabled by default and must fail independently from decoding.
FLEX	Optional bounded fragment reassembly adds a marked assembled copy while retaining original fragments.	Disabled by default; recording-backed live acceptance remains open.

Privacy and law: PDW cannot determine whether reception, storage, forwarding or publication is lawful in your location. Confirm authority, radio rules, privacy obligations, retention and publication requirements before use. Treat capcodes, aliases and decoded content as potentially identifying information.

Release status: Public Beta 2 passed automated x64/Win32 build, test, package, installer and Defender gates. It is intentionally unsigned, and physical SDR, SDR#, VB-CABLE, legacy receiver and complete UI acceptance remain incomplete. Do not describe it as signed, stable or fully hardware-validated.

2. Install and first start

Supported Windows boundary

Setup enforces Windows build 10586 as the technical API floor. Normal use requires a Windows 10 or Windows 11 edition/build still receiving Microsoft security servicing, or a serviced Windows Server 2016-or-newer release. Windows 7, 8 and 8.1 are outside the current Visual Studio 2026 and Windows SQLite support boundary.

Choose an architecture

Package	Choose it for	Native dependency rule
x64	Current 64-bit Windows, Windows audio, RTL-TCP and matching x64 receiver libraries.	All loaded native DLLs must be x64.
Win32	Bundled x86 RTL-SDR, historical slicer/receiver libraries or another known 32-bit integration.	All loaded native DLLs must be x86/Win32.

Architecture: A 32-bit DLL cannot load into x64 PDW, and a 64-bit DLL cannot load into Win32 PDW. Decoder, filter and configuration behaviour are shared between the builds.

Guided Setup

10. Verify the downloaded Setup checksum against the value published with Public Beta 2.
11. Run Setup and choose the destination folder. The default is under the current user's Local Programs folder.
12. On 64-bit Windows, choose x64 or Win32 compatibility. Setup selects Win32 automatically on 32-bit Windows.
13. For a new settings file, choose Standard PDW settings or the optional SDR# + VB-Audio Cable (Adelaide FLEX) profile.
14. Choose Start Menu, optional Desktop and optional delayed Windows-startup shortcuts.
15. Launch PDW and confirm the title/About identity and selected architecture before configuring hardware or outputs.

Unknown publisher: Public Beta 2 is intentionally unsigned. Windows or SmartScreen may identify an unknown publisher. Download only from the maintained release, verify the checksum, and do not bypass the warning for an unverified copy.

Upgrade and uninstall

16. Normal upgrades preserve PDW.INI, Capcode Directory/history database, receiver additions, WAV files, logs, recordings, queues and other operator-created data.
17. A pre-existing PDW.INI always wins; Setup does not silently apply a clean-install profile during an upgrade.
18. Architecture changes preserve the active RTL-SDR DLL to a verified recovery backup before replacing or removing the active slot.
19. Uninstall leaves configuration and operator data by default. Review and remove it manually only after an authorised backup.

Portable packages

The portable x64 and Win32 packages remain supported. Extract the complete package to a writable folder and keep the executable, PDW.INI, Capcode Directory/history database, receivers, WAV files and logs together. Do not run PDW from inside a ZIP file or mix files from different releases.

Important local data

Path / item	Purpose and handling
PDW.INI	Primary saved configuration beside PDW. Protect it as operator configuration.
pdw-history.sqlite3	Capcode Directory and, when enabled, message history. The directory remains usable with history capture off.
filters.ini.migrated	Recoverable backup created after a successful one-time legacy filter migration.
Wavfiles / Receivers	Optional alert audio and receiver packages. Native receiver DLLs must match PDW architecture.
Logfiles / Published / queues	May contain decoded or transformed content. Apply access, retention and disposal controls.
.pdwbackup	Password-encrypted configuration export that can include supported saved credentials.

3. Interface tour

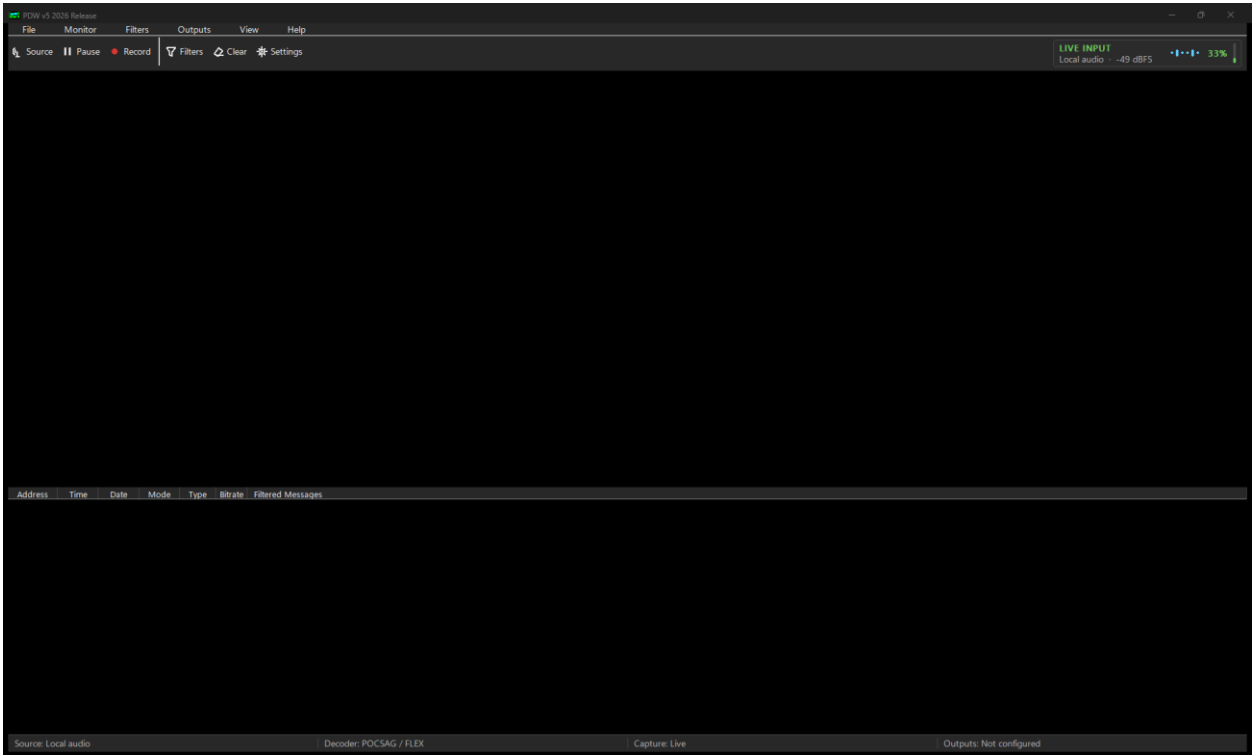


Figure 2. PDW v5 command bar, panes, live meter and status bar (content-free native capture).

Main window

Area	Use
Menus	File, Monitor, Filters, Outputs, View and Help organise the complete command set.
Command bar	Source, Pause, Record, Filters, Clear and Settings provide the primary daily actions.
Live input meter	Shows source state, source name, level/quality and real sample activity. It is not calibrated RF power.
Monitor pane	Upper pane for displayed decoded messages.
Filtered pane	Lower pane for rules whose Filter action is enabled and that are not rejected or Monitor Only.
Status bar	Reports source, decoder, capture state and output-health summary without displaying decoded text.

The six menus

Menu	Main tasks
File	Log, save/print copied data, configuration backup/restore and exit.
Monitor	Pause, decoder mode, input setup, radio/replay, statistics and volume.
Filters	Capcode Directory, filter controls, counters and legacy compatibility actions.
Outputs	Email, Apprise, transfer, publishing, data outputs, Delivery Health, dashboard and guarded channels.
View	Message history, display/column options, appearance and window tools.

Menu	Main tasks
Help	User guide, diagnostics, About, licence and project links.

Settings Center

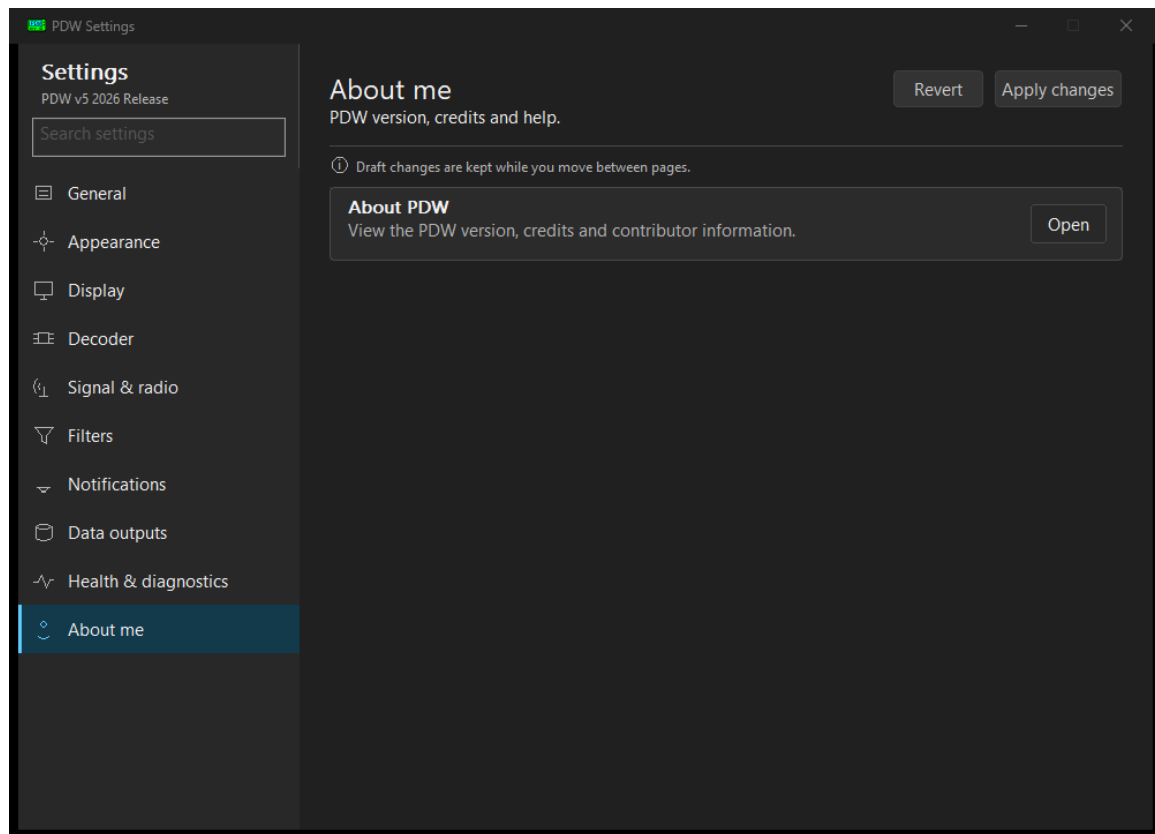


Figure 3. Modeless Settings Center navigation used by the v5 family.

Settings is modeless: monitoring continues while it is open. Draft changes are retained while moving between pages, and retained legacy editors return to the same page when closed. Opening Settings again focuses the existing window instead of creating a second copy.

Accessibility: Use clear message fonts and only the columns you need. Confirm keyboard navigation, High Contrast and 125-200% scaling on the intended computer before relying on the beta in an accessibility-sensitive workflow.

Development preview: consolidated Settings

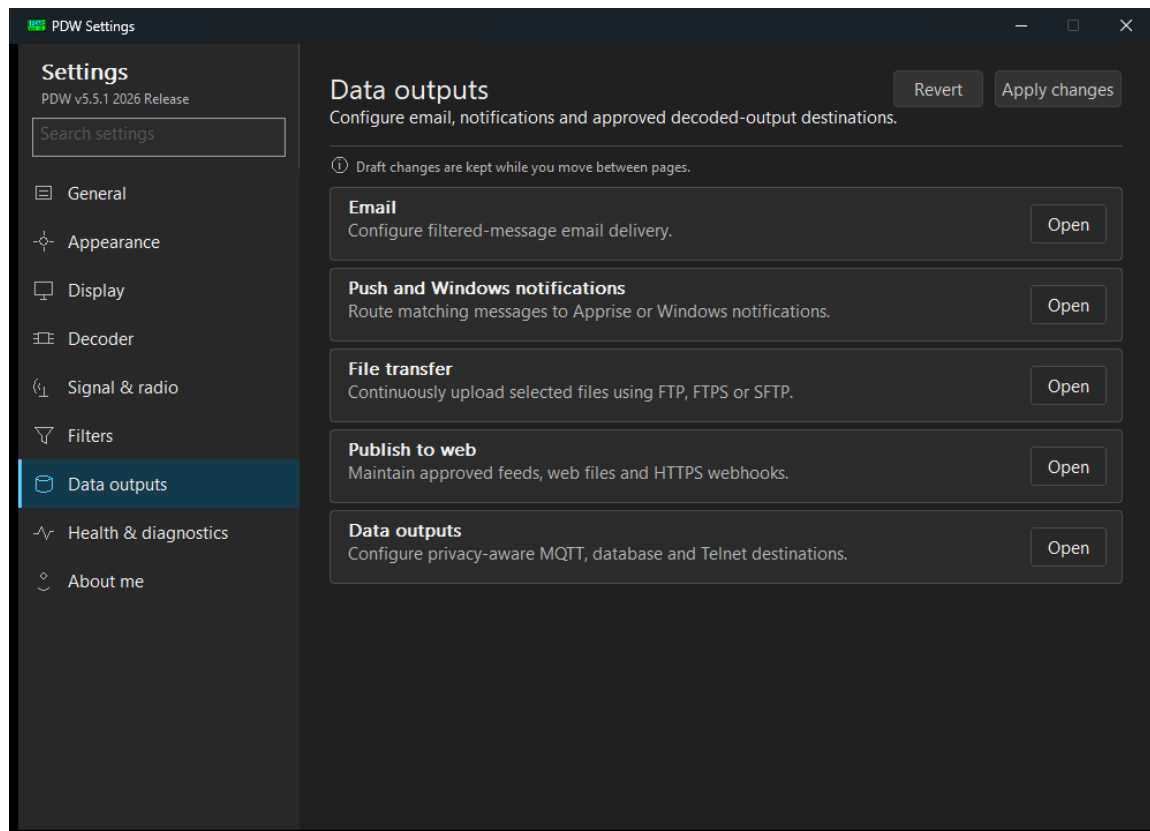


Figure 4. 11 August 2026 development preview of consolidated Data outputs navigation.

Development preview: The active source worktree consolidates duplicate Settings cards and groups email, push/Windows notifications, transfer, publishing and optional data outputs on one page. This screenshot is not a promise that the published Public Beta 2 menu count or wording has changed.

4. Choose a signal source

Source decision guide

Source	Use it when	Important boundary
Local audio	A receiver or virtual cable supplies audio.	Identityless profiles try legacy WinMM first; saved named endpoints open exact-ID WASAPI and fail closed if missing.
Serial / slicer / RS232	Known compatible legacy hardware is available.	Win32 remains the compatibility choice for x86-only integrations; obsolete VXD packages are not shipped.
RTL-TCP	A local or trusted-network rtl_tcp-compatible server supplies IQ.	Architecture-neutral TCP path; enhanced IQ filtering is optional and disabled by default.
Direct RTL-SDR	A compatible USB receiver is attached.	Win32 includes the reviewed x86 receiver pack; x64 needs a matching trusted x64 library or RTL-TCP.
Replay	A synthetic, redacted or licensed WAV/SigMF recording is available.	Runs through normal decoder functions and restores the previous live source.

Local audio and serial

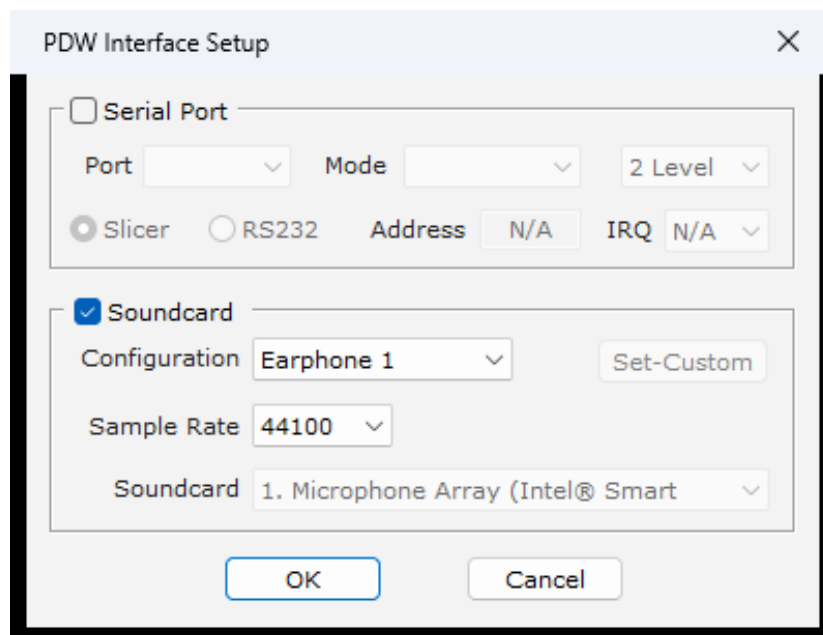


Figure 5. Legacy Input Setup remains available beside modern source controls.

20. Connect the intended receiver output, virtual cable or compatible serial interface before opening PDW.
21. Open Settings > Signal & radio and choose the current source or retained legacy input editor.
22. For audio, select the exact device and the closest supported configuration. Avoid persistent clipping.
23. For serial, select the exact COM port, Slicer or RS232 mode and the required two- or four-level path.
24. Select the matching protocol family and confirm valid synthetic/licensed traffic before saving automation settings.

SDR# + VB-Audio Cable (Adelaide FLEX)

The named profile is an explicit clean-install or operator-confirmed choice. SDR# sends unfiltered NFM audio to CABLE Input; PDW captures the matching CABLE Output recording endpoint. PDW stores the exact Windows endpoint identity and does not silently fall back to a microphone when that identity is missing or ambiguous.

- 25.SDR# and VB-Audio Cable are separate products; PDW does not install, license, tune or configure them.
- 26.The documented profile uses 148.8125 MHz, NFM, Filter Audio off, unmuted audio, FLEX 1600 and the audited decoder/slicer values.
- 27.Existing/portable users may preview Apply Adelaide FLEX. The confirmation defaults to No and creates a verified PDW.INI backup before an approved change.
- 28.The profile does not create, replace or delete Capcode Directory rules, message history, receivers or WAV files.

Hardware-unverified: Automated endpoint, fail-closed, profile and installer checks pass, but the exact physical SDR#/VB-CABLE workflow is not maintainer-validated for Public Beta 2. Describe it as configured or operator-reported until representative acceptance is recorded.

RTL-TCP and direct RTL-SDR

Radio and Signal Sources

Signal source: RTL-TCP compatible network receiver

Local inputs retain WinMM, serial and slicer behaviour; WASAPI remains an automatic fallback.

Receiver

Support package: RTL-SDR standard pack (RTL2832U, V3, V4, V4L) Add receiver... Open folder

RTL-TCP host: decoder-gateway.example.net Port: 1234 Connected device: 0

Frequency (Hz): 148337500 IQ sample rate: 1024000 Audio rate: 48000

☒ Automatic gain Manual gain (0.1 dB): 0 Correction (ppm): 1 NFM BW (Hz): 12500

Test selected source Receiver package is ready. Connect a compatible USB receiver and install WinUSB if needed.

Recording and replay

Record: C:\PDW-Training\Recordings\2026-08-10-test-signal.wav Browse... Start Stop

Replay: C:\PDW-Training\Recordings\synthetic-flex-training.wav Browse... Start Stop

WAV stores normalized decoder audio; SigMF stores normalized rf32_le samples.

Live signal and recording calibration

Calibrate replay...

Level 0% Noise 0% Clip 0.00% Eye 35% Signal 33% Errors 0/0 FLEX A:0 B:0 C:0 D:0

OK Cancel

Figure 6. RTL-TCP setup using fictional training values.

RTL-TCP validates the RTL0 header, configures the tuner, demodulates NFM and reconnects after a drop. Direct RTL-SDR lists devices and checks library architecture/API compatibility. Add receiver imports only a trusted matching-bitness librtlsdr package. PDW never launches Zadig or changes a USB driver automatically.

Record, replay and diagnose

- 29.WAV records normalized mono 16-bit samples; SigMF records rf32_le plus sample-rate metadata.
- 30.Replay accepts supported PCM8, PCM16 or float32 WAV and real float32 SigMF, resets timing, then restores the prior source.
- 31.The diagnostics view includes waveform, spectrum, waterfall, quality history, level, noise, clipping, eye opening and FLEX phase/error totals.
- 32.Calibration tests 1,000 threshold/centering/resync combinations and offers an operator-approved suggestion; it is not protocol-level proof.

Guarded multi-channel receivers

Outputs > Guarded multi-channel receivers can run up to four additional rtl_tcp or direct RTL-SDR channels. Each uses an isolated PDW worker process because the established decoders contain process-global state. Local history must be enabled, and each endpoint/device must be unique.

Isolation: Workers write channel-labelled events to shared local history but disable publishing, email, Apprise, data outputs, FTP, Telnet, dashboard and legacy logs. This release does not split one wideband IQ stream into several tuned channels.

5. Decode and improve a signal

Choose the decoder family

Use Monitor > Decoding mode or the Source/Mode route and select POCSAG/FLEX, ACARS, MOBITEK or ERMES. PDW monitors one protocol family at a time. Activity without valid messages often indicates the wrong family, tuning, bandwidth, level or source.

Improve reception in the right order

33. Confirm the source and protocol family.
34. Confirm frequency, receiver mode, sample rate and expected baud/rate.
35. Correct gross under-level or clipping and improve the antenna/signal path.
36. For SDR, adjust PPM, gain/AGC and NFM bandwidth using the same authorised recording or signal.
37. Compare the live meter, waveform, eye/phase information and corrected/uncorrectable totals.
38. Only then change custom thresholds or apply a calibration suggestion, keeping a reversible backup.

Optional RTL IQ conditioning

Enhanced IQ filtering and resampling applies only to RTL input. It uses a 60 dB windowed-sinc channel filter and anti-alias resampling before the existing decoder boundary. It is disabled by default; with it off, the established one-pole/averaging path remains. Compare the same lawful diagnostic recording before retaining the option.

Duplicate and message boundaries

Duplicate blocking, reject rules and protocol-specific display rules remain ahead of external output delivery. Optional output adapters receive an immutable event copy and cannot change the decoder, display or original filter result.

Public Beta 2 FLEX fragment assistance

Public Beta 2 can optionally assemble a complete non-group FLEX alpha/secure fragment chain. The feature is disabled by default. Original fragments continue through the established path, and one additional compact message marked [Joined FLEX] is emitted only after a bounded valid chain completes. Missing, conflicting, timed-out or capacity-rejected observations do not create a guessed joined message. FLEX Group Mode remains on its legacy path.

Live evidence: Automated fragment-order, replay, timeout, capacity and identity tests do not replace recording-backed live acceptance. Use only synthetic, redacted or licensed recordings.

Development preview: joined-only and Part X of Y

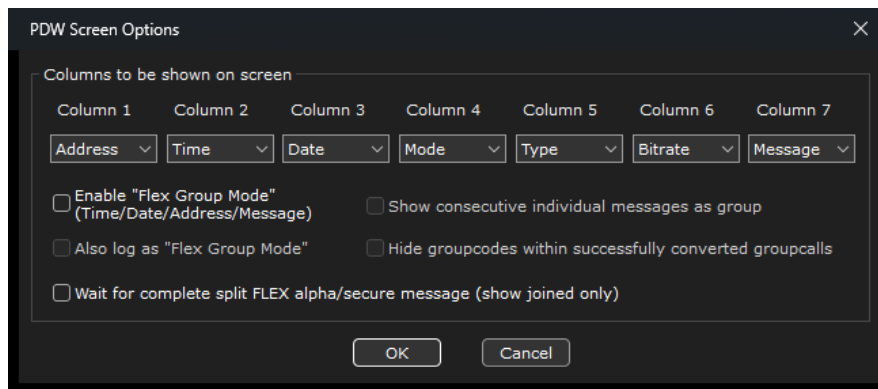


Figure 7. Development preview of the joined-only FLEX screen option.

The 11 August 2026 worktree changes the enabled FLEX option to hold valid fragments until one joined event enters display, filtering, logging and routing. It also recognises bounded explicit text markers such as Part 1 of 2 and can join 2-32 parts for one visible identity. These changes are development preview, not Public Beta 2 behaviour, and remain subject to build, test, package and recording-backed acceptance.

6. Understand decoded messages

Synthetic examples: The addresses, times, names and messages below are invented training data and do not identify a real person, aircraft or service.

Paging message fields

Address	Time	Mode	Type	Bitrate	Synthetic message
1234567	14:17:25	POCSAG-1	ALPHA	1200	TEST ONLY - maintenance drill complete
1765432	14:19:08	FLEX-A	ALPHA	1600	TRAINING - airport operations standby
1002048	14:22:31	FLEX-C	NUMERIC	3200	5550102

Address remains the raw decoded capcode/Riccode. Mode identifies protocol and function/phase. Type distinguishes alpha, numeric, tone-only or other protocol-specific content. Bitrate records the detected/selected rate. A current Capcode Directory name may be displayed beside the raw address but never replaces it.

ACARS and MOBITEK

ACARS records may include aircraft address/registration, flight, label, block/ack and message text. MOBITEK records may include network, base/mobile identity and protocol-specific message type. Keep only the columns and databases needed for an authorised workflow.

Capcode Directory

The local SQLite archive stores operator-maintained names, agencies, colours, notes and filter/routing rules. Protocol-specific mappings win over all-protocol mappings. The raw capcode remains available for audit and export.

Message history

Open View > Message history. History capture is disabled by default, and storing message text is a separate disabled-by-default choice. Retention may be set from 1 to 3650 days. The Capcode Directory remains available when history capture is off, and purging message history does not delete directory entries.

- 39.The default archive is pdw-history.sqlite3 beside PDW unless another path is selected.
- 40.The viewer searches and pages through 200-row displays without implying that only the visible page exists.
- 41.Corrupt or unrelated SQLite databases are refused unchanged; PDW does not silently repair or convert them.

Export message history to CSV

Export CSV writes every stored row matching the current Search, Protocol and Filtered controls, not just the displayed page. The UTF-8 file uses current enabled directory aliases at export time and leaves Message blank when text was not retained.

Received,Protocol,Capcode,Name,Agency,Type,Message,Filter

Exported data: A CSV is an unencrypted copy outside SQLite retention and purge controls. Formula-leading values are protected, but the file still requires authorised storage, sharing and deletion.

Local live dashboard

Outputs > Local live dashboard provides a read-only browser view and JSON API. It is disabled by default, requires message history, binds only to 127.0.0.1 and accepts only localhost/127.0.0.1 Host headers. It cannot accept remote connections or change PDW.

7. Build filters and route selected messages

[illegible]

Figure 8. Capcode Directory rule editor with explicit output destinations (content-free native capture).

One directory, three separate decisions

Decision	Control	Effect
Display	Display name, agency placement, colour	Adds operator-maintained context while retaining the raw capcode.
Pane	Filter, Monitor Only or Reject	Sends a match to the lower pane, keeps it in the upper pane, or suppresses it.
Delivery	Send to enabled outputs plus named destinations	Routes only to destinations that are also enabled/configured in Settings.

Create a rule

42. Open Filters > Capcode Directory and choose New.

43. Choose one protocol or Any protocol, then enter the exact capcode/address.
44. Add a synthetic display name, optional agency/service, colour and notes.
45. Choose capcode-only, one required keyword, two-to-ten required keywords joined with +, or exact whole-message matching.
46. Choose Filter, Monitor Only or Reject. Filter and Monitor Only are mutually exclusive.
47. Leave Send to enabled outputs off until each intended destination is separately approved, configured and tested.
48. Save or update. The live in-memory filter reloads immediately without replacing decoder logic.

Matching precedence

For the same capcode, PDW evaluates the most specific message condition first: exact whole-message, multi-keyword, single-keyword, then capcode-only. Keyword matching ignores case. A + expression requires every listed keyword in any order. Legacy & expressions remain supported for compatibility.

Per-rule outputs

A rule may select Email, Apprise, Publishing, MQTT, SQLite, MySQL/ODBC, Telnet and Windows notifications. Selecting a destination on the rule does not enable it. The destination and any required output-group/privacy acknowledgement must also be enabled in Settings. This two-step gate prevents a directory row from silently activating external delivery.

CSV import/export and legacy migration

49. Directory Import and Export use UTF-8 CSV. Invalid rows are rejected and reported.
50. Fresh packages do not include filters.ini. If an older readable file exists, PDW performs a transactional one-time merge into the directory.
51. After success, the original file is renamed to a unique .migrated recovery backup; a failed import leaves it unchanged.
52. Retire any scheduled legacy generator after migration and use the directory CSV controls for future bulk maintenance.

8. Log, display, copy and review

Display and columns

View/display settings control the seven message columns, colours, fonts, scrollbar and pane arrangement. Keep the raw address, time and protocol/type information needed for interpretation, but remove unused columns to improve readability. Long v5 messages can use the wider layout without changing decoder content.

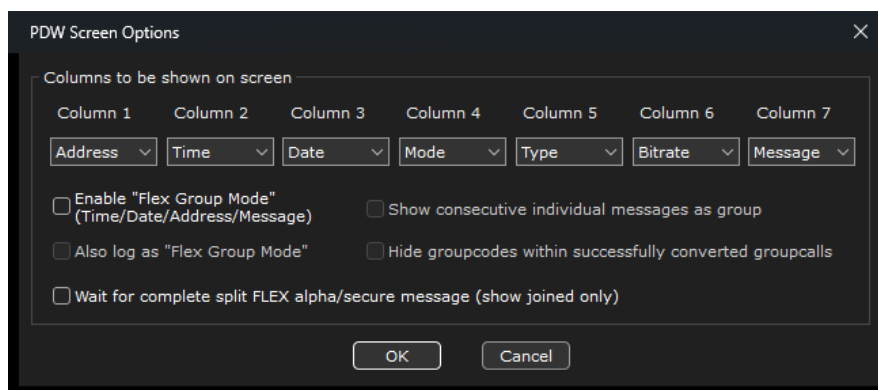


Figure 9. Current development Screen Options layout; the seven column selectors also exist in the v5 family.

Copy, save and print

53. Select decoded rows or text, then use Edit/Copy or the appropriate menu command.
54. Save copied data creates a separate operator-controlled file; it is not covered by message-history retention.
55. Print only to an approved device or secured PDF location and dispose of output under policy.

Main logfile

Logging is operator-controlled. Choose a protected location and confirm whether monitor and filtered content, labels and line layout meet the authorised purpose. A logfile may contain private decoded content even when external publishing has message omission enabled.

Separate filter CSV files

A directory rule can write the same matching row to up to three selected CSV paths using the shared subset of Capcode, Time, Date, Mode, Type, Bitrate and Message. These local files are independent from Message History CSV export and require their own access/retention controls.

Data boundary

Store/output	Contains	Retention boundary
Main logfile	Displayed/logged decoded records according to local settings.	Operator-controlled file.
Message history	Optional bounded SQLite events; text optional.	1-3650 day configured retention and purge.
History CSV	All stored rows matching current viewer filters.	Unencrypted export outside archive retention.
Published / queues	Transformed publishing copies and pending/dead-letter events.	Publishing profile and manual queue review.

Store/output	Contains	Retention boundary
Delivery Health	Destination names, counters, timestamps and sanitized outcomes.	Never stores capcodes or decoded message text.

9. Email and notifications

Two independent message gates

A Capcode Directory rule first decides whether a message is filtered/displayed and whether it selects an output. The email, Apprise or Windows-notification destination must then be enabled and configured in Settings. A rule never activates a destination by itself.

SMTP email

Figure 10. SMTP configuration using fictional, masked training values.

56. Use a current authenticated SMTP relay account and the provider's required TLS/port settings.
57. Test with a synthetic message and a non-production recipient before enabling matching-message delivery.
58. Do not place mail passwords or decoded traffic in screenshots, logs or issue reports.

Apprise filtered-message push

Apprise Notifications

☒ Enable Apprise

Apprise API notify URL

API username API password

Destination URL(s)

Separate multiple Apprise URLs with spaces or commas. Leave blank when using a configured /notify/{key} endpoint.

☐ Include filter details and decoded text (may appear on a phone lock screen)

Security

HTTPS and authenticated Basic access are required. Endpoint, destination URLs, and API credentials are stored in Windows Credential Manager and are never written to pdw.ini or logs.

Status

Apprise is disabled.

Apprise API v1.5.1 / Apprise v1.12.0

Send test notification Clear saved settings OK Cancel

Figure 11. Apprise notification settings with masked fictional values.

PDW connects to an operator-managed HTTPS Apprise API. Public Beta 2 sends only messages selected by the established filtered/output routing boundary. Monitor Only, rejected, duplicate and unselected messages are not pushed. Delivery uses a bounded background queue so an unavailable phone service does not block decoding.

59. Use HTTPS and authentication; do not expose an unauthenticated Apprise API to an untrusted network.

60. Keep decoded text off unless lock-screen exposure has been specifically approved.

61. Use Send test notification before Enable Apprise. Testing must use synthetic content.

Native Windows notifications

Windows notifications are a separate optional data-output destination. They load/register only when enabled or tested. Message text has its own opt-in because notifications may appear on the lock screen. Turning the adapter off leaves PDW's legacy tray behaviour unchanged.

Secrets: Supported destination credentials are held through Windows Credential Manager and the encrypted backup workflow. Never rely on an ordinary folder copy as a safe secret backup, and never publish an exported configuration file or its password.

10. Secure file transfer

The screenshot shows the 'Continuous File Transfer' dialog box with the following settings:

- Connection:**
 - Protocol: SFTP
 - Server: sftp.training-host.example
 - Port: 22
 - Username: pdw_training
 - Password: [masked]
 - Clear button
 - Remote folder: /public_html/pdw
 - ☒ Passive mode
 - SSH host key SHA256: SHA256:Vv8jQ6vXq1J6mAEoZCsIK8kUfjQHbVX1CjB6hEm9rl0
 - Text: SFTP encrypts the connection. Paste the provider's SHA256 SSH host-key fingerprint to prevent impersonation.
- Files to upload:**
 - List: C:\PDW-Training\Published\index.html, C:\PDW-Training\Published\messages.json, C:\PDW-Training\Published\messages.rss
 - Add files... button
 - Remove button
 - Text: Each file is uploaded using its existing filename.
- Schedule:**
 - ☒ Enable automatic uploads
 - Every 60 seconds (10 to 86400)
- Status:**
 - Text: FTP, FTPS and SFTP uploader is ready.
- Buttons:** Upload now, OK, Cancel

Figure 12. Secure file-transfer settings using fictional training values.

Supported protocols

Protocol	Protection	Required verification
FTP	No transport encryption	Use only inside an explicitly approved protected environment.
Explicit FTPS	TLS upgrade on the FTP control connection	Windows-trusted certificate and hostname validation.
Implicit FTPS	TLS from connection start	Windows-trusted certificate and hostname validation.
SFTP	SSH file transfer with password authentication	Exact operator/provider-verified SHA-256 host-key fingerprint.

Configure an upload

62. Create and approve the remote account/folder outside PDW.
63. Choose the secure protocol, host, port, username, remote folder and local source file.
64. For SFTP, obtain the exact SHA-256 host-key fingerprint through a trusted channel. For FTPS, use the correct certificate hostname.

65. Enter the password through PDW's credential control and run the configuration test.

66. Confirm the synthetic/local test file arrives in the intended folder before enabling continuous transfer.

67. Set a reasonable interval and review sanitized FileTransfer.log outcomes.

Fail closed: Do not bypass certificate or host-key validation to make a test pass. A changed SFTP host key or certificate error must be investigated through an independent trusted channel.

11. Publish to a website or webhook

Figure 13. Publishing settings using fictional training values.

Start disabled and minimise data

Publishing is disabled by default and requires an operator acknowledgement of permissions, privacy, radio and publication responsibility. Start with filtered-only delivery, address masking and message omission. Those transforms affect only the published copy; they do not rewrite the local display, logs, email or Apprise source event.

Static publishing

PDW can generate JSON, JSONL, RSS, Atom and responsive HTML. Generate to a private local folder first, inspect only synthetic output, then use the separately configured transfer path if remote upload is approved.

HTTPS webhook

68. Use an HTTPS endpoint with reviewed retention, region, access and account controls.
69. Bearer authentication and HMAC signatures are optional tools; the receiving workflow must actually validate them.
70. Use the stable event identifier for idempotency where the receiver supports it.
71. Run Test webhook with a configuration-only synthetic object before enabling live delivery.

Queue and failure behaviour

72.Delivery runs away from the capture/decoder thread.

73.Transient failures retry with bounded exponential backoff; repeated failures move to DeadLetter for review.

74.Pending/dead-letter payloads may contain transformed decoded data and require the same protection as the destination.

Third-party storage: A webhook or hosting provider may retain requests, headers, files and event history. Do not send live decoded content until the service and workflow are approved for that data.

12. Data outputs and delivery health

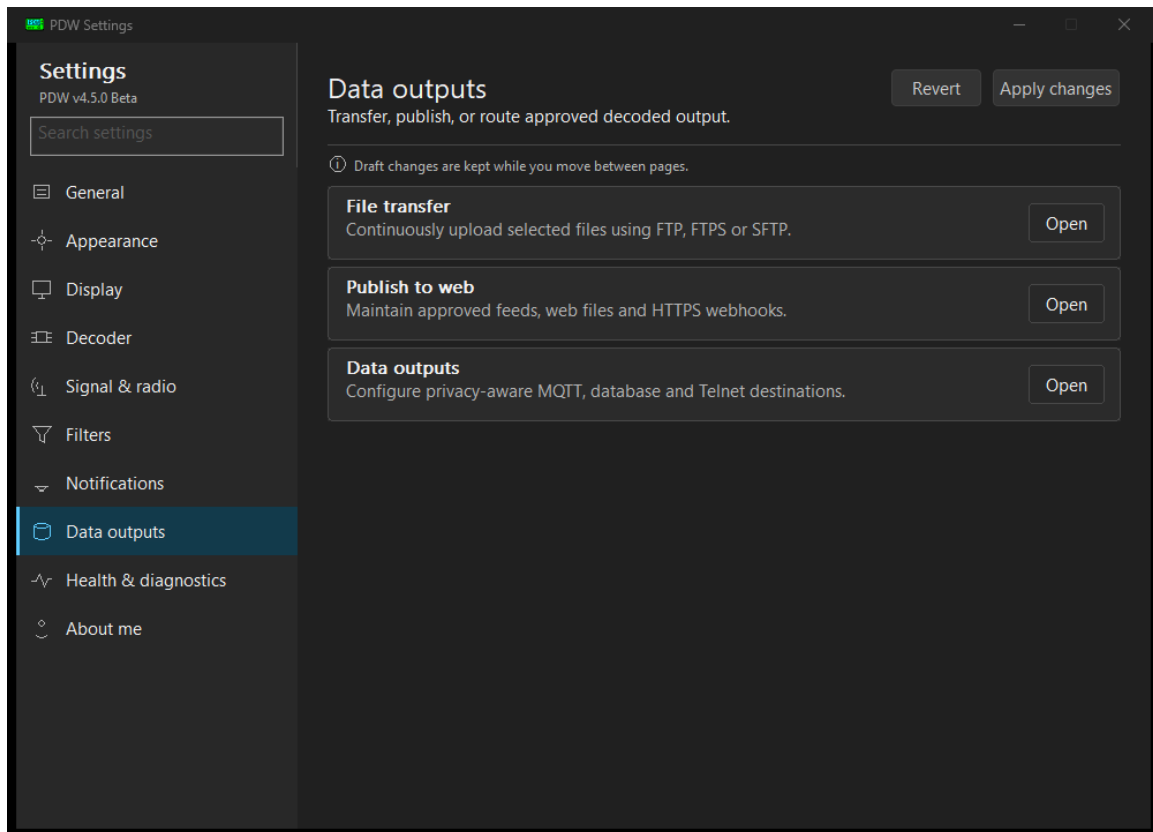


Figure 14. Data outputs area in the v5 Settings family.

Privacy-aware optional adapters

Optional adapters receive an immutable copy after PDW's established filtering and deduplication boundary. The output group and each adapter are disabled by default. Enabling the group requires an operator acknowledgement; filtered-only delivery, address masking and message omission are the privacy-safe defaults.

Adapter	Use	Important boundary
MQTT	Publish JSON to a broker/topic.	mqtt verifies certificate/hostname; plain mqtt needs explicit trusted-local opt-in; QoS 0 and no retain.
SQLite	Write prepared, indexed local rows.	Creates only when enabled/tested; event ID primary key makes replays idempotent.
MySQL / ODBC	Write through a current Windows ODBC DSN.	Driver, database and TLS policy are external; PDW uses prepared inserts.
Telnet JSON	Read-only one-JSON-object-per-line stream.	Loopback by default; non-loopback is explicit and unencrypted; no commands are accepted.
Windows	Native per-user notifications.	Message text separately opt-in because lock-screen display is possible.

Two-step routing gate

A live message reaches an adapter only when the matching directory rule selects that destination and Settings has both the output group and adapter enabled/configured. Legacy non-directory-backed rules retain their documented compatibility behaviour. Configuration tests use synthetic events; SQLite/MySQL tests may deliberately write one identified test row.

Queue and failure isolation

Each adapter runs on a bounded worker queue of 500 events. An adapter error is recorded for health reporting and never blocks decoding or prevents another selected adapter from receiving its copy.

Delivery Health

Outputs > Delivery Health shows per-destination enabled state, successes, failures, drops, consecutive failures, last outcome and bounded recent operational history. Alerts may flash the taskbar or play a warning after a configured threshold without stealing focus.

Content-free health: Delivery Health stores destination names, timestamps, counters and sanitized summaries only. It never stores capcodes or decoded message text and cannot alter routing, retry, capture or decoding.

13. Daily operation and troubleshooting

Start-of-session checklist

75. Confirm the PDW version, architecture, instance/folder and selected profile.
76. Confirm the intended source, decoder family and live meter without exposing private content.
77. Check system date/time, disk space and message-history retention state.
78. Confirm the correct Capcode Directory rules and whether Filter versus external routing is intended.
79. Review Delivery Health and any publishing/transfer dead-letter items before enabling new delivery.
80. Run a synthetic filtered and unfiltered test after a material configuration change.

Encrypted configuration backup

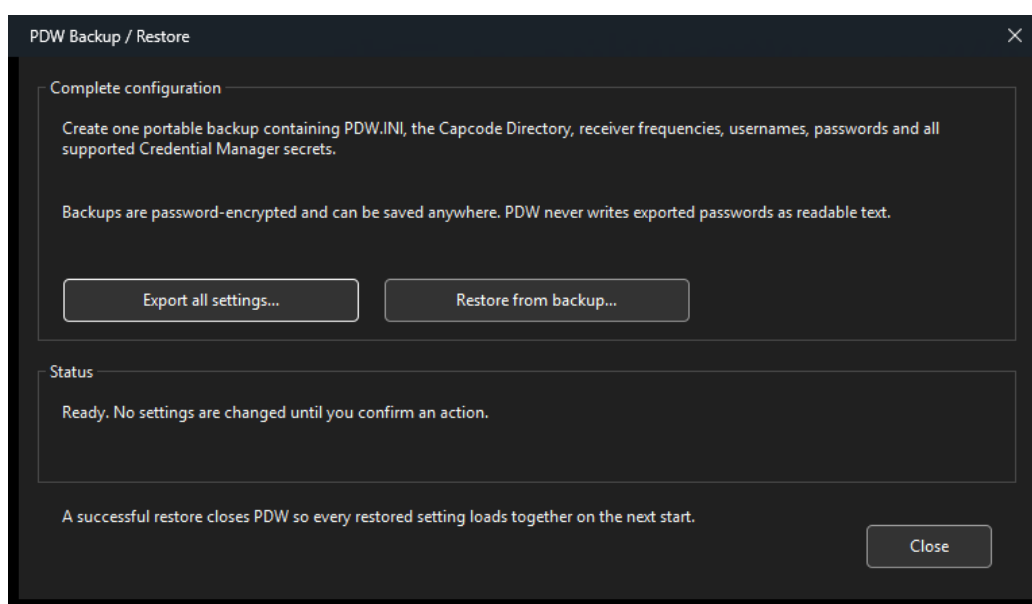


Figure 15. Password-encrypted Backup / Restore dialog (content-free native capture).

Settings > General > Backup / Restore creates one .pdwbackup file containing saved PDW.INI values, the complete Capcode Directory/filter rule set and supported Credential Manager records. AES-256-GCM protects the file with a user-chosen password derived using PBKDF2-HMAC-SHA256. PDW cannot recover a forgotten password.

81. Restore authenticates and validates the complete file before confirmation and transactional replacement.
82. A successful restore closes PDW so every restored setting loads together on restart.
83. The backup does not contain decoded messages, logs, recordings, output queues, website files, receiver DLLs or program binaries.

Backup handling: A .pdwbackup can contain usernames and secrets. Store the file and its password separately, restrict access, test restore in an isolated copy, and securely delete superseded backups under policy.

Troubleshooting

Symptom	Checks
No live meter	Confirm exact source, Windows permission, cable/device, saved endpoint identity, COM/receiver selection

Symptom	Checks
	and source test.
Activity, no messages	Confirm protocol family, frequency, PPM, rate, bandwidth, gain/level, antenna and clipping.
Named endpoint missing	Reconnect/enable the exact VB-Cable endpoint or deliberately choose another input; PDW should not capture the default microphone.
RTL-SDR not listed	Close other SDR software, confirm the exact USB interface/driver, and use a matching-bitness trusted receiver library.
Rule does not match	Check protocol/capcode, exact versus + keywords, precedence, enabled state, Reject and Monitor Only.
No external delivery	Confirm both the rule destination and Settings destination/group are enabled, configured and passing a synthetic test.
History CSV seems short	Check Search, Protocol and Filtered controls; blank Message means text was not retained.
Queue grows	Pause exposure if needed, correct the destination, review Delivery Health/dead letter, then retest with synthetic data.
Settings layout clips	Record exact beta, architecture, Windows build, DPI, theme and content-free screenshot; do not attach operator configuration.

Safe acceptance

84. Use an isolated training folder or test installation.
85. Use synthetic, redacted or explicitly licensed recordings and directory rows.
86. Exercise one known matching and one non-matching event through display, history and local outputs.
87. Use disposable external test destinations and keep message text omitted.
88. Confirm wrong TLS, authentication and SFTP host-key conditions fail closed.
89. Record the exact beta, architecture and Windows environment without endpoint IDs, credentials or private traffic.
90. Remove disposable credentials, exports and test rows after acceptance.

14. Reference

Quick menu reference

Menu	Use
File	Local logs/copies, configuration backup and exit.
Monitor	Source, pause, mode, input/radio/replay, statistics and volume.
Filters	Capcode Directory, rule behaviour, counters and migration compatibility.
Outputs	Email, notification, transfer, publishing, adapters, health, dashboard and extra channels.
View	Message history, columns, display and appearance.
Help	Manual, diagnostics, About, licence and project resources.

Safe defaults

Capability	Default
Network and external outputs	Disabled until deliberately configured and enabled.
Message history	Disabled; storing message text separately disabled.
Local dashboard	Disabled; loopback-only when enabled.
Guarded extra channels	Disabled; each slot must be enabled and started.
FLEX fragment assembly	Disabled in Public Beta 2.
Enhanced RTL IQ filtering	Disabled; established RTL signal path retained.
Adelaide FLEX profile	Standard settings remain the default; named profile is explicit and reversible.

Architecture and file reference

Item	Reference
Executable	PDW v5.5.1 2026 Release.exe
x64 role	Current Windows and matching x64 libraries; RTL-TCP is architecture-neutral.
Win32 role	Bundled x86 RTL-SDR and legacy 32-bit integrations.
Configuration	PDW.INI beside the executable.
Directory/history	pdw-history.sqlite3 by default.
Configuration export	Password-encrypted .pdwbackup.
History export	UTF-8 CSV outside archive retention.

Glossary

Term	Meaning
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Term	Meaning
Capcode / Riccode	Raw paging address used for matching; an operator alias never replaces it.
Filtered pane	Lower pane selected by a rule's Filter action; independent from external delivery.
Monitor Only	Keeps a match in the upper pane and does not make it a lower-pane filtered result.
WASAPI	Modern Windows audio capture API used for exact saved endpoints and as a controlled fallback.
WinMM	Established Windows multimedia audio path retained for compatibility.
RTL-TCP	TCP service supplying tuner IQ samples; no decoded message text is sent to it.
SigMF	Signal Metadata Format used here for real float32 diagnostic samples plus metadata.
FTPS / SFTP	TLS-protected FTP / SSH File Transfer Protocol; they use different verification models.
Dead letter	Repeatedly failed queued delivery retained for authorised operator review.
Development preview	Current source work that is not yet part of the published beta contract.

Credits and licence

PDW retains credits to Jason Petty, Peter Hunt, the 2013 open-source release and later contributors, including the current modernization work credited by the application. The application and source identify GNU General Public License version 3 or later. Review the supplied License and THIRD_PARTY_NOTICES.md for complete terms and component notices.

Manual maintenance: This edition targets the immutable v5.5.1 Public Beta 2 release and separately identifies 11 August 2026 development previews. Re-check version, menus, security guidance, third-party service requirements and hardware acceptance before packaging it with a later beta or stable release.